Ethan Mahajan

Mechanical Engineering Student

emahajan7@gatech.edu | Dacula, GA | ethan-mahajan | EthanMahajan

Professional Summary

- Mechanical engineering student with experience in CAD design, additive manufacturing, electromechanical prototyping, and freelance engineering consulting. Completed 20+ client projects while independently designing and building a high-speed CoreXY 3D printer.

Education

Georgia Institute of Technology, Mechanical Engineering — Atlanta, GA
- GPA: 4.00 — Aug 2026 – May 2029
- Relevant Coursework: Principles & Applications of Engineering Materials; Engineering Graphics

Mill Creek High School, STEM — Hoschton, GA
- Weighted GPA: 4.63; SAT: 1540 (Top 1% Nationally) — Aug 2022 – May 2026
- AP Scholar with Distinction; National Honor Society
- Physics and Math Scholar Award (1 of 700 Students)
- Relevant Coursework: AP Physics; AP Calculus; AP Chemistry

Skills

CAD & Design: Fusion 360, Solidworks, Onshape, FreeCAD, Technical Drawings, Design for Manufacturing

Manufacturing & Fabrication: FDM 3D Printing, Rapid Prototyping, Mechanical Assembly, Woodworking, Metalworking, Basic Metrology

Electronics & Hardware: PCB Design, Soldering, Wiring, Crimping, Multimeter Diagnostics, CAN Bus, Connector Assembly

Machinery & Tools: FDM 3D Printers, Laser Engravers, Manual & CNC Milling, & Wood Saws

Software & Systems: Microsoft Office, Google Workspace, Github, Windows, Linux, MacOS, Klipper, Java, Python, & Markdown

Communication & Project Work: Technical & Client Communication, Team Collaboration, Project Management & Documentation, Online Tutoring, Lesson Planning, Time Management

Projects

FDM 3D Printer Design, Prototyping, & Revising — June 2024 – July 2026

Created a high-speed CoreXY 3D printer by overhauling an older machine.
- Designed a frame and motion system in CAD to meet mechanical and thermal specifications.
- Verified component tolerances and clearances before fabrication.
- Fabricated a prototype using metalworking machines and additive manufacturing.
- Configured Kalico Klipper firmware for high-speed, quality 3D prints.
- Installed CAN-bus toolhead electronics to compartmentalize for easier maintenance.
- Used vibration analysis and input shaping to improve rigidity, reliability, and print speed.

Air Purifier — July 2026 – present

Smart DIY air purifier that reuses a 120 mm computer case fan and responds to room particulate-matter levels.
- Designed a control concept for automatic fan-speed adjustment and filter replacement detection.
- Completed electrical schematics in KiCad; PCB placement and testing are coming soon.

Laser-Engraved Chess Board Aug 2026 – present

In-progress Invention Studio project combining CAD, laser engraving, and planned CNC-machined metal pieces.

- Developed a quickly manufacturable board design and planned low-center-of-mass pieces for stability.
- Fabricated the laser-engraved playing surface and personalized reverse; CNC-machined metal pieces remain planned.

Experience

Fiverr, Freelance CAD Designer & Manufacturing Consultant — Remote

Completed 20+ custom engineering contracts involving CAD design, reverse engineering, and small-scale manufacturing. Feb 2024 – Oct 2025, 1 year 9 months

- Maintained a 4.6-star client rating across completed engagements.
- Designed and iterated an accessibility device, including a tie-tying assistance tool.
- Reverse-engineered a broken replacement component from reference materials and technical drawings.
- Manufactured a low-volume production run of 30 tailored tablet holders.

Shape Machine, Social Media & Web Design Contributor — Atlanta, GA

Work with an early-stage startup developing shape-computation tools and supporting public-facing digital communication. Sept 2026 – present, 2 months

- Launched Shape Machine's digital presence through social-media outreach.
- Collaborated with an interdisciplinary team to publish the Shape Computation Lab website.
- Contribute GitHub documentation and website fixes through the team's enterprise repository.

Wyzant, Freelance Private Tutor — Remote

Work directly with students of different ages to schedule lessons, prepare instructional materials, and tailor tutoring to each student's needs. May 2026 – Aug 2026, 4 months

- Tutored approximately 100 hours while maintaining a 5-star rating across approximately 70 reviews.
- Schedule lessons directly with students and prepare materials that clarify each lesson's subject matter.
- Developed online tutoring, time-management, lesson-planning, and student-communication skills.

Mathnasium of Dacula, Mathematics Instructor — Dacula, GA

Instructed students from foundational arithmetic through AP Calculus while adapting explanations to individual learning speeds. Aug 2024 – Aug 2025, 1 year 1 month

- Instructed up to five students simultaneously in a fast-paced learning environment.
- Translated complex mathematical concepts into clear, step-by-step explanations.
- Supported student confidence and academic progress across diverse grade levels.

Mercor & Handshake AI, AI Data Trainer & Labeler — Remote

Analyze AI-generated artifacts and author prompts to provide structured feedback on model quality. May 2026 – present, 6 months

Involvement

Technology Student Association, Treasurer (TSA) — Hoschton, GA
Aug 2024 – May 2026

Managed club finances, supported engineering competition activities, and helped coordinate member communication.

- Managed club finances and structural budgeting alongside faculty advisors.
- Organized community fundraising initiatives, including a local movie-night event.
- Organized and delivered presentations about upcoming events and member opportunities.
- Competed at the state level as part of a biotechnology team.

Osborne Middle School, Volunteer Tutor — Aug 2025 – May 2026

Provided volunteer tutoring support to middle-school students.

Languages

English
Native

German
Conversational

Interests

Advanced Manufacturing & Research: Additive Manufacturing, Rapid Prototyping, Robotics & Electromechanical Systems, Hardware & Electronics, Experimental Design, Applied Engineering Research

Systems, Software & AI: Programming, Operating Systems (Windows, macOS, Linux), Engineering Software, Local & Agentic AI Experimentation

Design, Craft & Personalization: Personalized Product Design, Everyday Problem Solving, Woodworking, Culinary Experimentation

Outdoors, Strategy & Learning: Hiking & Outdoor Exploration, Chess, Continuous Technical Learning